

Random Network Routing Evaluation

Autonomous Networks Research Group (ANRG)
University of Southern California

Contents

1	Evaluation Setup	2
2	Makespan vs. Density	2
3	Additional Metrics vs. Density (Large DAG)	3
4	Best Routing per Density and Scheduler	5
4.1	Small DAG (8 tasks)	5
4.2	Large DAG (30 tasks)	6
5	Full Results Tables	6
5.1	L150 (avg. degree 24.0) — Small DAG (8 tasks)	6
5.2	L200 (avg. degree 17.1) — Small DAG (8 tasks)	7
5.3	L250 (avg. degree 12.4) — Small DAG (8 tasks)	7
5.4	L300 (avg. degree 8.7) — Small DAG (8 tasks)	7
5.5	L350 (avg. degree 6.7) — Small DAG (8 tasks)	8
5.6	L400 (avg. degree 5.8) — Small DAG (8 tasks)	8
5.7	L500 (avg. degree 3.7) — Small DAG (8 tasks)	8
5.8	L150 (avg. degree 24.0) — Large DAG (30 tasks)	9
5.9	L200 (avg. degree 17.1) — Large DAG (30 tasks)	9
5.10	L250 (avg. degree 12.4) — Large DAG (30 tasks)	9
5.11	L300 (avg. degree 8.7) — Large DAG (30 tasks)	10
5.12	L350 (avg. degree 6.7) — Large DAG (30 tasks)	10
5.13	L400 (avg. degree 5.8) — Large DAG (30 tasks)	10
5.14	L500 (avg. degree 3.7) — Large DAG (30 tasks)	11
6	Analysis	11
6.1	HEFT-1 Dominance Is Robust Across All Densities	11
6.2	SH vs. S: Density-Dependent Reversal	11
6.3	Hops, Peak Link Util., and Transfer Duration	11
6.4	Reproducing These Results	12

1 Evaluation Setup

Does the relative performance ranking of HEFT, HEFT-1, and HEFT-2 hold on random graphs? We place **50 nodes** uniformly at random in an $L \times L$ square and create bidirectional links between all pairs within $R = 80$ m. Varying L controls network density.

Parameter	Value
Nodes	50, positions uniform random in $[0, L]^2$
Communication range	$R = 80$ m (<code>csma_bianchi</code> PHY rate from distance)
Topology seed	42 (fixed per density level)
Seeds per combo	30 (seeds 1–30)
Schedulers	HEFT (calibrated), HEFT-1, HEFT-2
Routing schemes	W, S, SH, GO, GS, GSD, GSD-D
DAGs	Small (8 tasks, fork-join) and Large (30 tasks, pipeline)
Interference	<code>csma_bianchi</code> , 802.11ax 5 GHz 20 MHz

Table 1: Evaluation parameters.

Values reported as mean $\pm$ 95 % confidence interval over 30 seeds. Additional metrics: mean hops (path length), peak link utilization (P-LU), mean transfer duration (XD).

Level	Side L (m)	Links	Avg. degree
L150	150	601	24.0
L200	200	427	17.1
L250	250	309	12.4
L300	300	217	8.7
L350	350	168	6.7
L400	400	146	5.8
L500	500	92	3.7

Table 2: Topology statistics (seed 42).

2 Makespan vs. Density

Each point is the best-routing mean makespan $\pm$ 95 % CI over 30 seeds ($t_{29} = 2.045$).

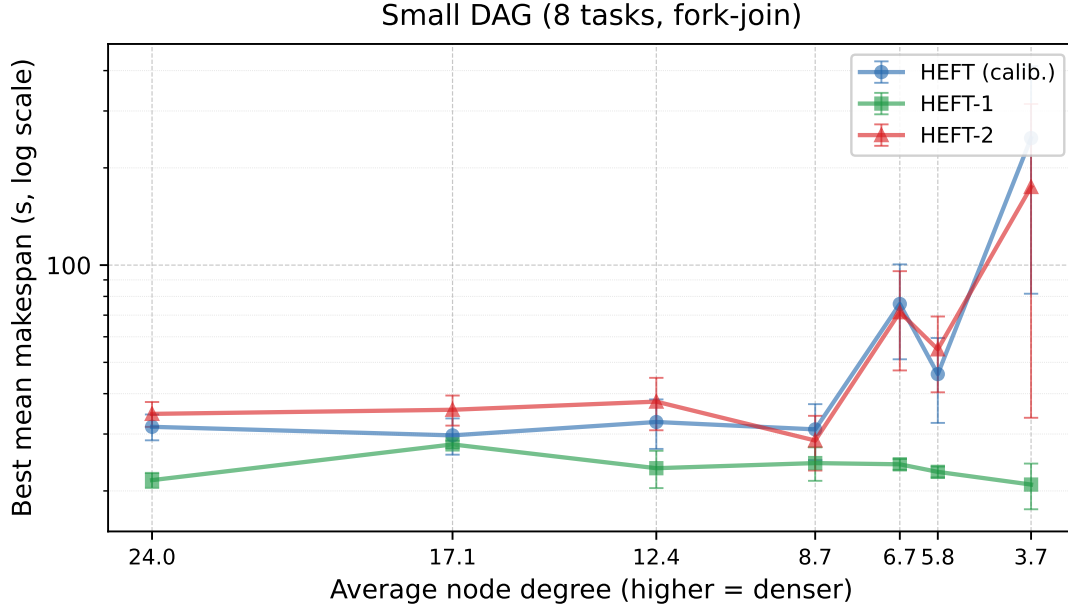


Figure 1: Best-routing makespan vs. avg degree — Small DAG (8 tasks, fork-join). Error bars show 95 % CI over 30 seeds.

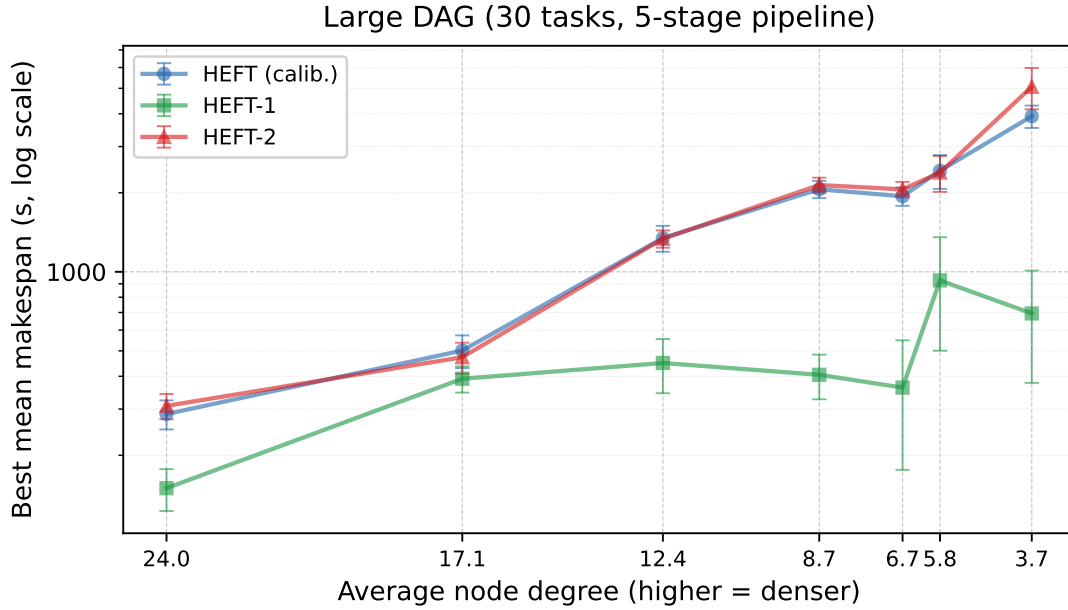


Figure 2: Best-routing makespan vs. avg degree — Large DAG (30 tasks, 5-stage pipeline). Error bars show 95 % CI over 30 seeds.

3 Additional Metrics vs. Density (Large DAG)

Computed at the best routing scheme per (density, scheduler) cell.

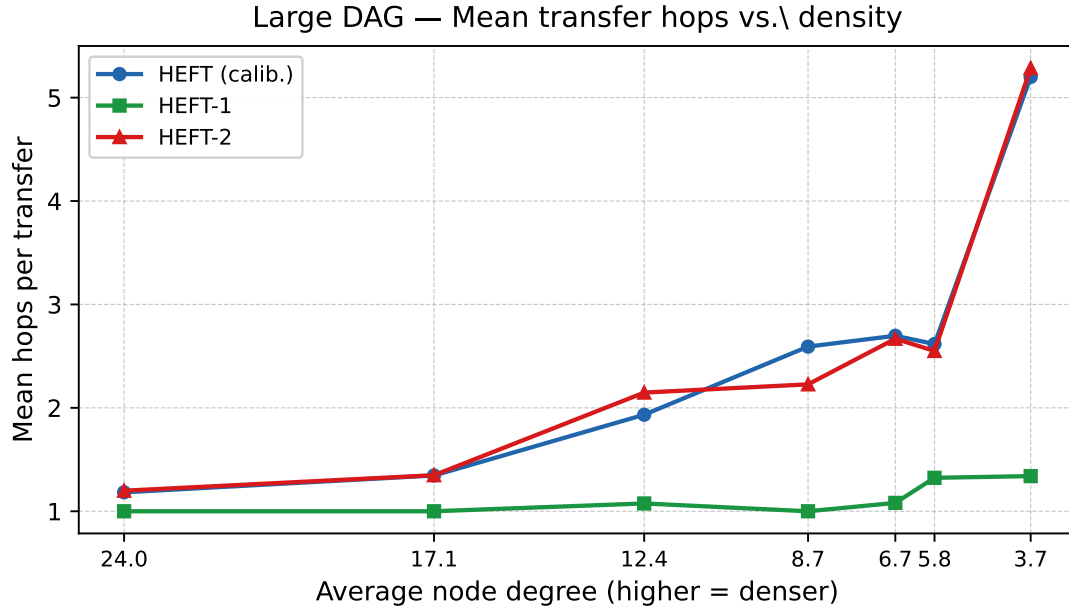


Figure 3: Mean hops per transfer. Values > 1 indicate multi-hop paths.

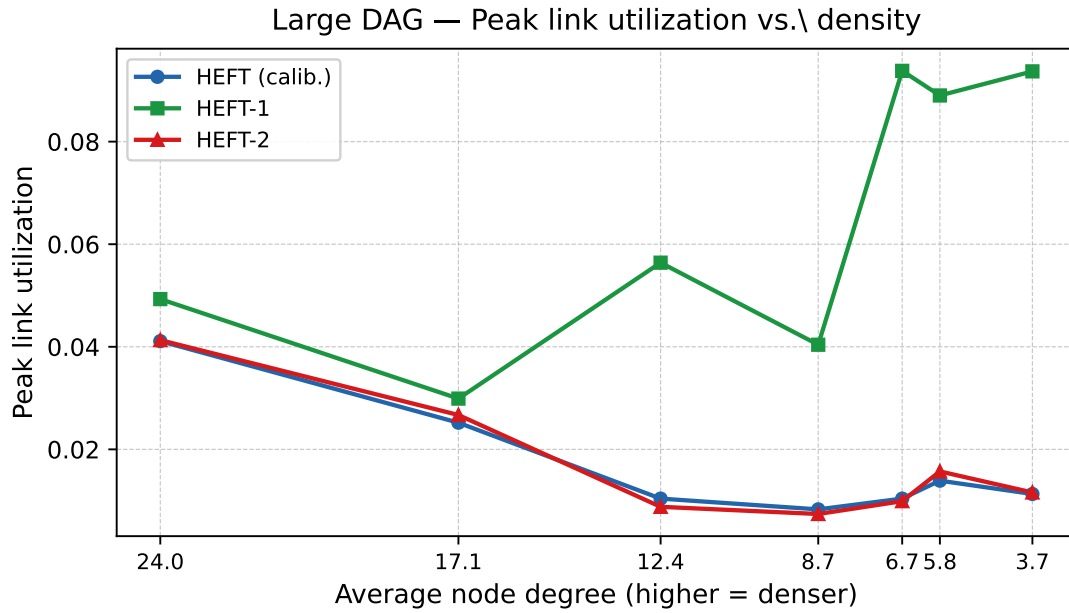


Figure 4: Peak link utilization (max over all links). High values indicate congestion hotspots.

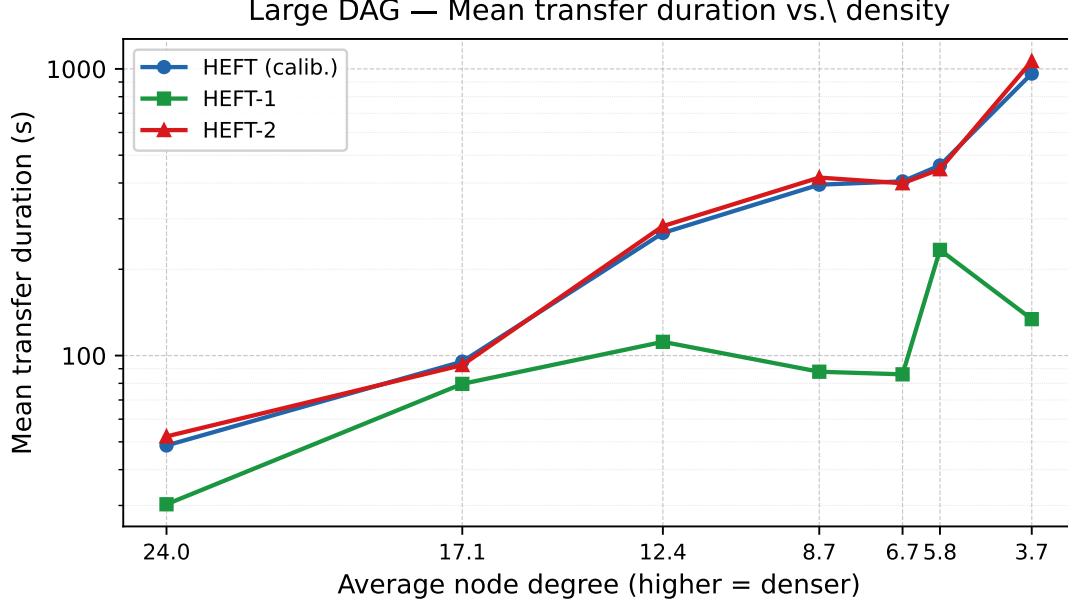


Figure 5: Mean transfer duration (s, log scale). Tracks makespan closely.

4 Best Routing per Density and Scheduler

4.1 Small DAG (8 tasks)

Dens.	Deg.	HEFT (calib.)			HEFT-1			HEFT-2		
		Rt	ms (s)	H/PLU	Rt	ms (s)	H/PLU	Rt	ms (s)	H/PLU
L150	24.0	S	31.584 ±2.900	1.1/0.20	S	21.594 ±1.156	1.0/0.19	S	34.633 ±3.060	1.1/0.21
L200	17.1	SH	29.712 ±3.818	1.0/0.20	S	27.861 ±0.748	1.0/0.16	SH	35.664 ±3.807	1.0/0.22
L250	12.4	S	32.688 ±5.706	1.1/0.19	S	23.513 ±3.108	1.0/0.17	S	37.796 ±6.982	1.2/0.18
L300	8.7	SH	31.009 ±6.110	1.3/0.18	SH	24.388 ±2.895	1.0/0.18	SH	28.641 ±5.516	1.2/0.17
L350	6.7	SH	75.816 ±24.712	1.8/0.17	SH	24.179 ±0.933	1.0/0.22	SH	71.486 ±24.280	1.7/0.17
L400	5.8	SH	45.980 ±13.503	1.3/0.18	S	22.909 ±0.881	1.0/0.24	SH	54.861 ±14.472	1.4/0.20
L500	3.7	S	247.261±165.707	2.1/0.17	S	20.926±3.378	1.0/0.29	GS	174.510±140.827	1.8/0.21

Table 3: Small DAG (8 tasks) — best routing per (density, scheduler). ms = mean ± 95% CI; H = mean hops; PLU = peak link util. **Bold green**: overall best.

4.2 Large DAG (30 tasks)

Dens.	Deg.	HEFT (calib.)			HEFT-1			HEFT-2		
		Rt	ms (s)	H/PLU	Rt	ms (s)	H/PLU	Rt	ms (s)	H/PLU
L150	24.0	SH 287.323	±36.446	1.2/0.04	SH 149.899	±27.286	1.0/0.05	SH 308.494	±33.945	1.2/0.04
L200	17.1	SH 501.174	±71.407	1.3/0.03	SH 391.400	±44.749	1.0/0.03	SH 472.856	±64.069	1.3/0.03
L250	12.4	SH 1344.549	±152.451	1.9/0.01	S 449.477	±104.817	1.1/0.06	S 1336.595	±101.142	2.1/0.01
L300	8.7	S 2063.615	±155.323	2.6/0.01	SH 405.276	±78.324	1.0/0.04	SH 2140.430	±142.610	2.2/0.01
L350	6.7	S 1939.103	±155.199	2.7/0.01	SH 362.181	±186.120	1.1/0.09	S 2060.780	±139.772	2.7/0.01
L400	5.8	SH 2426.675	±357.120	2.6/0.01	W 927.487	±427.012	1.3/0.09	SH 2388.566	±373.325	2.6/0.02
L500	3.7	S 3914.460	±383.147	5.2/0.01	GSD-D 694.194	±316.752	1.3/0.09	GSD-D 5068.567	±909.731	5.3/0.01

Table 4: Large DAG (30 tasks) — best routing per (density, scheduler). ms = mean $\pm$ 95 % CI; H = mean hops; PLU = peak link util. **Bold green**: overall best.

5 Full Results Tables

Mean $\pm$ 95 % CI makespan (s), mean hops, peak link util over 30 seeds. **Bold green**: overall best for density+DAG. **Red**: worst per scheduler column.

5.1 L150 (avg. degree 24.0) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	494.056±147.762	4.0	0.017	43.651±7.933	2.2	0.108	520.163±163.541	4.3	0.018
S	31.584±2.900	1.1	0.198	21.594 ±1.156	1.0	0.187	34.633±3.060	1.1	0.211
SH	36.964±3.438	1.0	0.208	21.956±1.275	1.0	0.187	35.639±3.540	1.0	0.204
GO	536.541±98.678	3.6	0.044	242.145±6.110	3.8	0.059	612.818±91.968	3.6	0.036
GS	589.876±67.393	3.7	0.040	286.759 ±30.755	3.4	0.059	562.633±70.565	3.8	0.042
GSD	640.136 ±107.023	4.3	0.038	286.083±41.503	2.0	0.056	639.593 ±109.650	4.4	0.040
GSD-D	414.070±32.423	4.7	0.058	144.089±12.548	2.0	0.095	409.654±44.804	4.9	0.063

Table 5: L150 ($L = 150$ m) — Small DAG (8 tasks). ms = mean $\pm$ 95 % CI; H = mean hops; PLU = peak link util.

5.2 L200 (avg. degree 17.1) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	517.000±198.909	3.8	0.076	43.892±9.260	1.9	0.127	460.683±185.125	3.6	0.083
S	38.313±7.874	1.3	0.158	27.861 ±0.748	1.0	0.164	38.134±7.369	1.3	0.171
SH	29.712±3.818	1.0	0.200	28.909±0.895	1.0	0.190	35.664±3.807	1.0	0.217
GO	872.239±464.202	3.5	0.066	411.368±149.125	3.1	0.068	1184.579 ±553.872	3.5	0.062
GS	1088.416 ±527.152	3.3	0.060	489.974 ±175.122	3.1	0.057	888.119±461.644	3.4	0.067
GSD	990.931±496.353	3.0	0.052	483.804±175.641	2.2	0.053	936.577±453.634	4.7	0.053
GSD-D	459.646±270.976	3.9	0.115	229.148±79.280	1.8	0.109	830.506±402.973	3.8	0.091

Table 6: L200 ($L = 200$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.3 L250 (avg. degree 12.4) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	609.587 ±284.649	2.8	0.116	29.521±6.402	1.1	0.165	546.569±275.384	2.7	0.116
S	32.688±5.706	1.1	0.192	23.513 ±3.108	1.0	0.170	37.796±6.982	1.2	0.180
SH	41.687±7.461	1.3	0.172	25.843±3.481	1.0	0.174	38.534±7.328	1.2	0.170
GO	596.181±318.328	2.9	0.086	106.264±36.084	2.5	0.111	405.795±262.923	2.7	0.101
GS	250.967±106.538	2.7	0.106	91.428±30.504	2.4	0.119	314.147±107.516	2.8	0.093
GSD	468.916±201.021	2.6	0.088	118.211 ±36.289	1.1	0.115	629.508 ±220.720	3.1	0.075
GSD-D	190.333±55.839	2.5	0.116	84.703±28.674	1.1	0.132	209.977±66.194	3.1	0.101

Table 7: L250 ($L = 250$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.4 L300 (avg. degree 8.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	847.135±358.602	3.4	0.106	28.512±3.223	1.1	0.178	720.078±348.734	3.1	0.120
S	34.708±6.641	1.4	0.166	26.982±3.235	1.0	0.181	33.490±6.498	1.3	0.168
SH	31.009±6.110	1.3	0.181	24.388 ±2.895	1.0	0.175	28.641±5.516	1.2	0.171
GO	1067.400 ±476.239	2.5	0.072	151.303 ±51.901	2.3	0.112	730.413±421.413	2.3	0.083
GS	916.827±439.461	2.6	0.071	129.153±50.966	2.1	0.122	576.451±332.842	2.3	0.083
GSD	361.737±241.696	2.1	0.116	109.158±48.063	1.2	0.133	871.740 ±315.716	3.8	0.073
GSD-D	476.092±295.972	2.4	0.127	116.086±43.200	1.3	0.156	475.722±262.206	2.7	0.119

Table 8: L300 ($L = 300$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.5 L350 (avg. degree 6.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	692.048±384.056	2.8	0.111	24.745±2.495	1.1	0.233	985.791±415.467	3.4	0.091
S	414.203±310.178	2.2	0.141	25.020±1.971	1.1	0.211	331.440±284.289	1.9	0.153
SH	75.816±24.712	1.8	0.169	24.179±0.933	1.0	0.217	71.486±24.280	1.7	0.167
GO	1115.620±539.034	2.3	0.094	40.942±8.600	1.6	0.189	1312.776±554.026	2.5	0.084
GS	1015.965±527.695	2.2	0.099	52.288±8.288	1.8	0.170	625.956±447.208	1.9	0.115
GSD	336.893±334.757	1.9	0.123	54.115±8.107	1.2	0.168	1215.382±547.496	3.2	0.088
GSD-D	1113.997±539.157	3.0	0.103	39.763±8.833	1.1	0.223	917.628±512.704	2.7	0.113

Table 9: L350 ($L = 350$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.6 L400 (avg. degree 5.8) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	837.123±404.511	3.1	0.138	29.321±3.320	1.3	0.209	689.413±384.756	2.7	0.144
S	615.176±371.411	2.6	0.139	22.909±0.881	1.0	0.238	615.634±371.306	2.6	0.149
SH	45.980±13.503	1.3	0.181	23.741±0.876	1.0	0.231	54.861±14.472	1.4	0.203
GO	503.103±329.977	2.7	0.108	207.313±63.658	2.1	0.118	512.725±328.218	2.8	0.104
GS	440.730±305.939	2.6	0.106	255.294±51.647	2.5	0.086	384.038±277.460	2.6	0.119
GSD	437.806±306.482	2.8	0.122	243.985±58.167	1.3	0.086	592.820±345.125	3.2	0.109
GSD-D	423.879±308.347	2.3	0.138	117.097±43.150	1.1	0.182	433.288±307.046	2.7	0.137

Table 10: L400 ($L = 400$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.7 L500 (avg. degree 3.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	356.336±189.844	2.9	0.168	21.669±3.063	1.0	0.282	210.881±154.391	2.0	0.197
S	247.261±165.707	2.1	0.174	20.926±3.378	1.0	0.286	392.711±195.288	2.9	0.149
SH	501.809±205.282	3.4	0.121	23.151±3.280	1.0	0.297	465.443±202.948	3.2	0.131
GO	501.795±205.287	3.5	0.134	22.127±2.757	1.0	0.277	465.438±202.949	3.5	0.135
GS	429.050±199.641	3.2	0.167	21.384±3.119	1.0	0.282	174.510±140.827	1.8	0.211
GSD	392.711±195.288	3.1	0.149	32.377±8.185	1.0	0.208	356.363±189.838	2.8	0.141
GSD-D	318.912±183.442	2.6	0.173	21.863±3.235	1.0	0.312	318.765±183.475	2.6	0.198

Table 11: L500 ($L = 500$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.8 L150 (avg. degree 24.0) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1530.829±83.415	5.5	0.007	542.056±116.043	2.5	0.020	1504.689±87.893	5.6	0.007
S	715.431±104.235	1.7	0.020	198.784±37.917	1.2	0.047	675.535±115.115	1.7	0.022
SH	287.323±36.446	1.2	0.041	149.899±27.286	1.0	0.049	308.494±33.945	1.2	0.041
GO	3540.427±252.412	3.8	0.008	2117.429±284.512	3.5	0.021	3383.308±250.924	3.8	0.009
GS	3567.396±274.503	3.9	0.008	2501.810±258.116	3.7	0.016	3589.212±263.342	3.8	0.008
GSD	3256.273±157.867	5.5	0.008	2192.774±282.031	2.5	0.018	3309.441±144.260	5.5	0.008
GSD-D	2601.980±109.213	5.4	0.010	1805.307±203.539	2.6	0.021	2568.484±104.218	5.5	0.011

Table 12: L150 ($L = 150$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.9 L200 (avg. degree 17.1) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1943.409±145.184	5.7	0.007	1079.533±183.910	2.5	0.016	1822.756±142.774	5.4	0.007
S	831.126±43.227	1.8	0.016	410.010±51.479	1.0	0.030	860.765±57.385	1.8	0.015
SH	501.174±71.407	1.3	0.025	391.400±44.749	1.0	0.030	472.856±64.069	1.3	0.027
GO	4049.688±224.275	3.7	0.010	3425.021±470.992	3.6	0.015	4131.985±170.785	3.7	0.010
GS	3973.898±304.565	3.7	0.010	4263.847±803.378	3.5	0.011	3671.203±273.076	3.7	0.010
GSD	3531.402±252.443	5.6	0.010	3902.399±746.826	2.5	0.013	3652.316±274.661	5.7	0.010
GSD-D	3131.900±321.521	5.5	0.013	2520.026±176.308	2.3	0.018	2981.828±313.006	5.7	0.013

Table 13: L200 ($L = 200$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.10 L250 (avg. degree 12.4) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1812.396±116.015	4.8	0.009	626.690±103.974	1.9	0.032	1852.661±105.025	4.7	0.009
S	1357.554±97.448	2.1	0.008	449.477±104.817	1.1	0.056	1336.595±101.142	2.1	0.009
SH	1344.549±152.451	1.9	0.010	559.928±103.137	1.0	0.044	1361.136±146.432	1.9	0.010
GO	3063.776±164.294	3.7	0.009	1262.239±110.987	3.1	0.029	2986.991±152.055	3.7	0.008
GS	3048.733±185.128	3.4	0.007	1820.238±372.158	3.0	0.025	3023.702±175.828	3.5	0.008
GSD	2530.877±142.307	4.8	0.008	1835.618±329.462	2.0	0.019	2602.391±124.457	4.8	0.008
GSD-D	1951.639±117.022	4.7	0.011	1226.291±249.919	1.9	0.029	1924.656±119.640	4.7	0.011

Table 14: L250 ($L = 250$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.11 L300 (avg. degree 8.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	2639.992±236.721	4.9	0.006	809.067±239.281	1.9	0.047	2726.267±229.055	5.5	0.006
S	2063.615±155.323	2.6	0.008	428.149±92.782	1.1	0.041	2156.510±115.283	2.7	0.007
SH	2233.182±135.497	2.4	0.007	405.276 ±78.324	1.0	0.040	2140.430±142.610	2.2	0.007
GO	2810.638±156.588	4.0	0.008	1078.668±138.360	1.9	0.033	2859.602 ±159.704	4.0	0.008
GS	2822.218 ±178.021	3.9	0.009	1027.168±106.392	1.9	0.040	2807.280±201.110	3.9	0.009
GSD	2769.135±91.596	5.2	0.008	1471.472 ±228.230	1.9	0.022	2725.680±103.706	5.1	0.008
GSD-D	2740.669±304.335	5.6	0.009	998.452±18.802	1.8	0.035	2500.433±346.116	5.2	0.010

Table 15: L300 ($L = 300$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.12 L350 (avg. degree 6.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	2262.481±116.134	4.3	0.009	627.582±289.139	1.3	0.093	2378.291±98.254	4.3	0.009
S	1939.103±155.199	2.7	0.010	767.926±257.048	1.3	0.071	2060.780±139.772	2.7	0.010
SH	2142.480±199.353	2.5	0.011	362.181 ±186.120	1.1	0.094	2155.497±214.279	2.6	0.011
GO	2577.160±319.352	3.6	0.010	880.890±288.824	1.7	0.050	2406.193±284.951	3.4	0.011
GS	2426.125±261.843	3.9	0.010	1185.292 ±327.594	1.8	0.041	2483.959 ±243.395	3.7	0.012
GSD	2625.803 ±279.153	3.9	0.010	1028.508±274.115	1.3	0.036	2454.205±272.643	4.0	0.010
GSD-D	2575.997±359.438	4.4	0.011	891.407±280.561	1.3	0.050	2353.488±339.628	4.1	0.011

Table 16: L350 ($L = 350$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.13 L400 (avg. degree 5.8) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	2844.226±310.613	3.6	0.006	927.487 ±427.012	1.3	0.089	2980.537±292.734	4.4	0.006
S	2947.258±303.632	3.8	0.007	1080.936±474.566	1.1	0.067	2919.019±294.283	4.3	0.007
SH	2426.675±357.120	2.6	0.014	1252.065±494.741	1.1	0.064	2388.566±373.325	2.6	0.016
GO	3331.649 ±103.409	4.1	0.009	1229.818±509.775	1.6	0.072	3336.711 ±111.771	4.1	0.008
GS	3154.020±118.777	4.7	0.009	1611.463±532.376	1.6	0.063	3239.241±131.545	4.5	0.009
GSD	3209.028±160.826	3.9	0.008	1621.605 ±576.469	1.3	0.035	3065.852±138.373	4.4	0.007
GSD-D	3070.476±255.271	3.8	0.008	1202.449±504.778	1.3	0.068	3205.536±226.138	4.3	0.007

Table 17: L400 ($L = 400$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5.14 L500 (avg. degree 3.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	5494.190±1013.775	5.6	0.010	906.925±350.604	1.3	0.097	5732.305±862.441	5.0	0.011
S	3914.460±383.147	5.2	0.011	792.757±339.975	1.0	0.108	6166.377±1062.026	4.9	0.011
SH	5924.170±1038.246	5.0	0.011	1035.847±363.267	1.1	0.091	6870.982±962.830	4.7	0.011
GO	4771.168±737.807	5.3	0.012	923.603±359.152	1.2	0.100	5773.163±910.336	4.6	0.013
GS	4544.688±643.535	5.5	0.011	860.593±352.926	1.2	0.104	5767.751±1000.618	4.8	0.012
GSD	5560.900±891.405	4.7	0.013	938.632±355.988	1.2	0.086	5386.756±720.862	4.7	0.013
GSD-D	5212.339±875.525	5.1	0.012	694.194±316.752	1.3	0.094	5068.567±909.731	5.3	0.012

Table 18: L500 ($L = 500$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

6 Analysis

6.1 HEFT-1 Dominance Is Robust Across All Densities

HEFT-1 wins at every density level for both DAGs. For the large DAG, the advantage over calibrated HEFT ranges from $1.9\times$ at L150 to $5.6\times$ at L500. The standard deviations under HEFT and HEFT-2 are large (often $> 50\%$ of the mean) because csma_bianchi interference is highly sensitive to the exact traffic pattern, which varies across seeds. HEFT-1’s std is much smaller: co-location eliminates most transfers, leaving only compute variance.

6.2 SH vs. S: Density-Dependent Reversal

Under HEFT-1, SH beats S at L150–L300 for the large DAG (mean-hops = 1.0 for both, confirming co-location). The routing tie-break wins come from how each scheme selects the single hop: SH prefers the nearest (lowest-loss) link, reducing PHY-layer retransmission and thus interference variance. At sparse topologies (L400–L500), fewer 1-hop options exist and S regains the lead by picking the highest-bandwidth available link.

6.3 Hops, Peak Link Util., and Transfer Duration

The additional metrics confirm that routing choice under csma_bianchi is primarily about *which* 1-hop link is used, not multi-hop avoidance:

- Mean hops is ≈ 1.0 for HEFT-1 across all density levels, confirming that co-located tasks transfer directly to adjacent nodes.
- Calibrated HEFT and HEFT-2 use 2–4 hops at high density (L150–L200), where tasks are spread across many nodes. Multi-hop paths under csma_bianchi amplify interference—each relay adds an active transmitter—explaining the large makespan gap.
- Peak link utilization under HEFT-1 is low (< 0.06) because co-located tasks do not use links; the few transfers that occur are short and non-overlapping. Under HEFT/HEFT-2 peak link util can exceed 0.9, indicating severe bottleneck congestion.

6.4 Reproducing These Results

```
cd ncsim/  
python run_random_eval.py          # ~3500 s (8820 runs)  
python compute_interference_metrics.py  # reads traces, ~3 min  
python gen_random_network_report.py    # generates tex + plots  
cd docs/ && pdflatex random_network_results.tex
```